



8 The curve C has equation $y = \tanh x$ for $x \geq 0$.

(a) Sketch C and state the equation of the asymptote.

[2]

(b) By considering a suitable set of N rectangles of unit width, use your sketch to show that

$$\sum_{r=1}^N \tanh r > \ln(\cosh N). \quad [3]$$



(ii) Find the exact value of $\pi \int_{\frac{5}{4}}^{\sqrt{2}} \frac{u^2}{u^2 - 1} du$. You need not simplify your answer. [3]

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